Colour & Optics

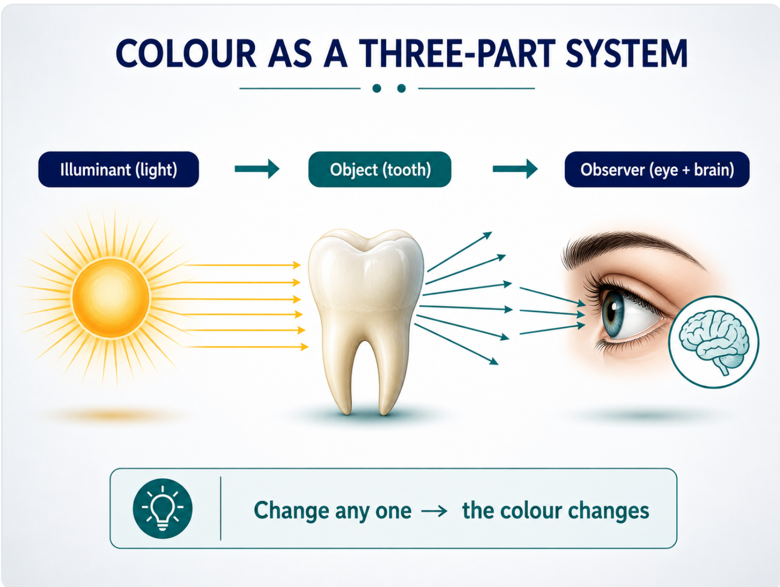
Student handout — colour is not a pigment; it is what the material does with light.

Revision aid, not a full transcript. Organised on the structure → property → performance spine.

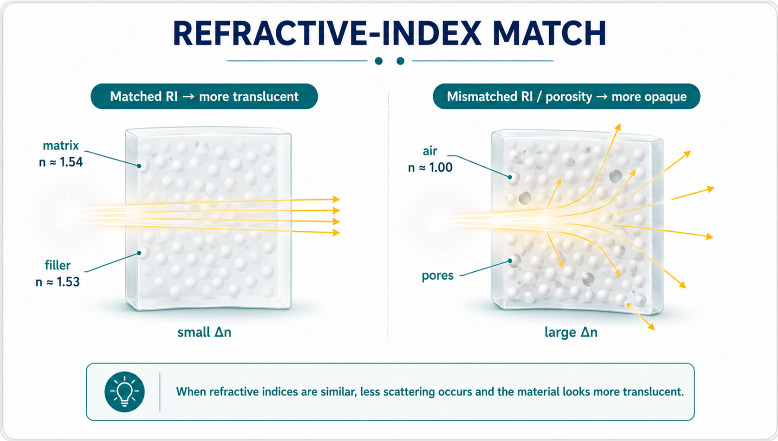
STRUCTURE

Light & the material's microstructure

- Colour is a three-part event: **light source + object + observer**. Change any one and the perceived colour changes.
- The eye samples the **visible spectrum (~380–780 nm)** with three cone types (S/M/L) → trichromatic then opponent processing.
- **Refractive index ($n = c/v$)** = how much a material slows and bends light. Mismatch in n at an interface → light bends and scatters.
- Key values: **enamel ≈ 1.63**, **dentine ≈ 1.54**, hydroxyapatite ≈ 1.62–1.65, water 1.33, air ≈ 1.0; Bis-GMA 1.54, silica ≈ 1.53, TEGDMA 1.46.
- In composites, matching **filler and matrix n** → translucency; air/water-filled pores (n ≈ 1.0, e.g. a white-spot lesion) scatter strongly → opaque white.



Colour = source × object × observer.



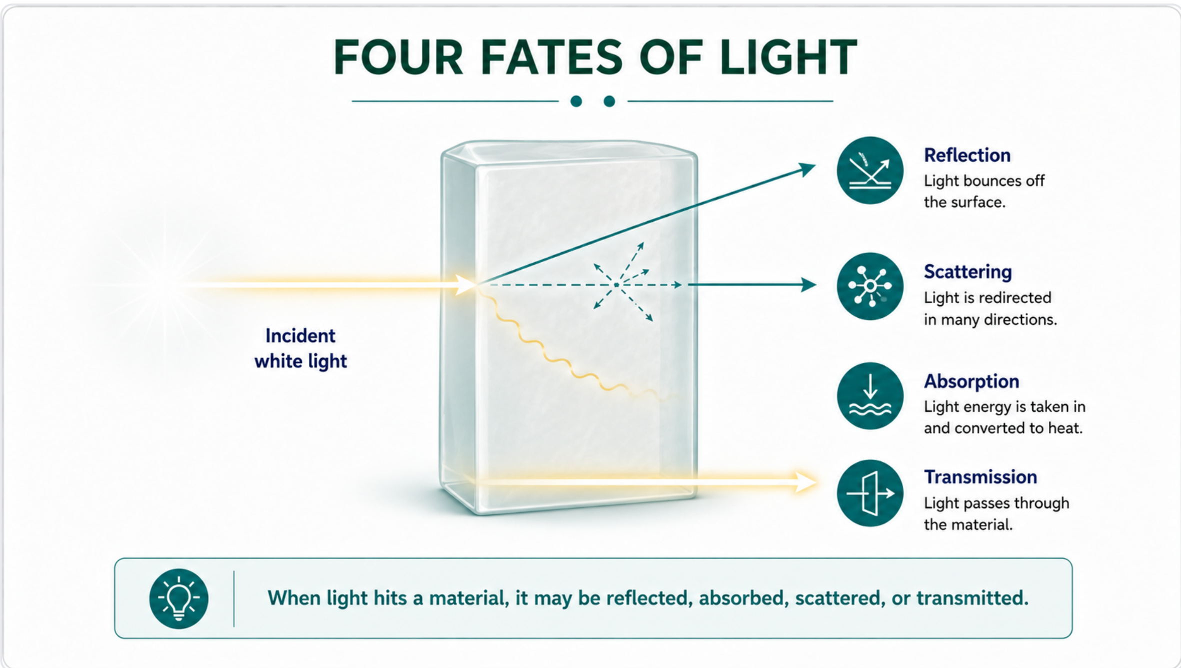
Filler–matrix RI match → translucent; mismatch (or pores) → scatter/opaque.

PROPERTY

What light does & the three attributes

The four fates of light

- **Reflection** (specular = glossy surface; diffuse = scattered) and **absorption** act at the surface.
- **Scattering** and **transmission** act in the body of the material — they govern translucency.
- The mix of these four determines the final appearance.

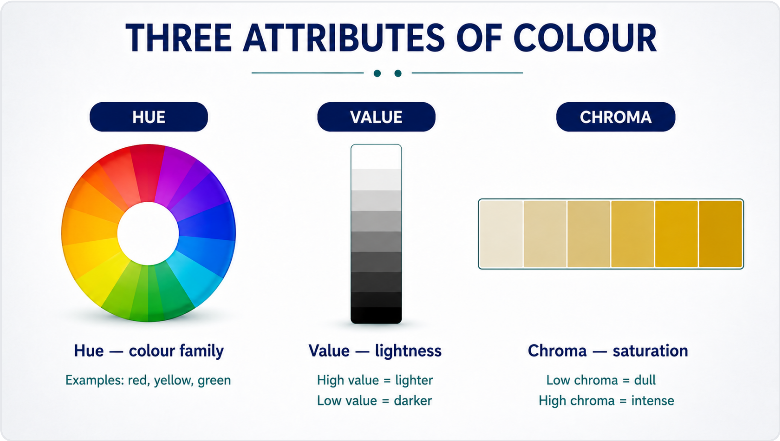


Four fates of light: reflection & absorption (surface), scattering & transmission (body).

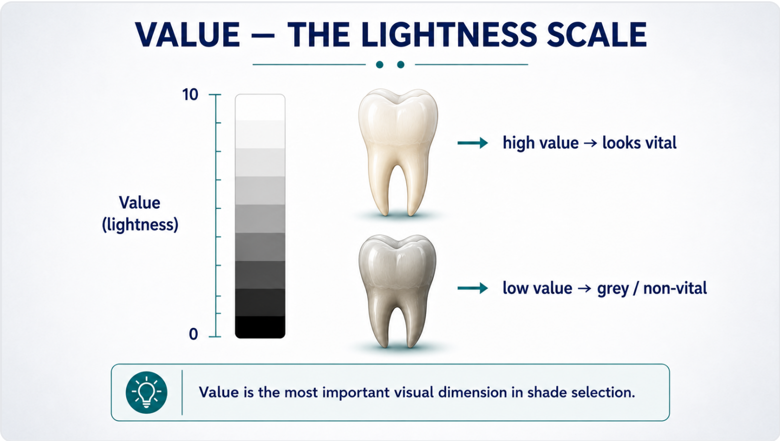
Hue · Value · Chroma

- **Hue** — the colour family (teeth sit in yellow / yellow-red; Vita families A–D).
- **Value** — lightness (0–10). **Value dominates** perceived match: squint to judge it; when unsure "go lighter".

- **Chroma** — colour intensity/saturation (highest cervically, lowest incisally); tends to move inversely with value.



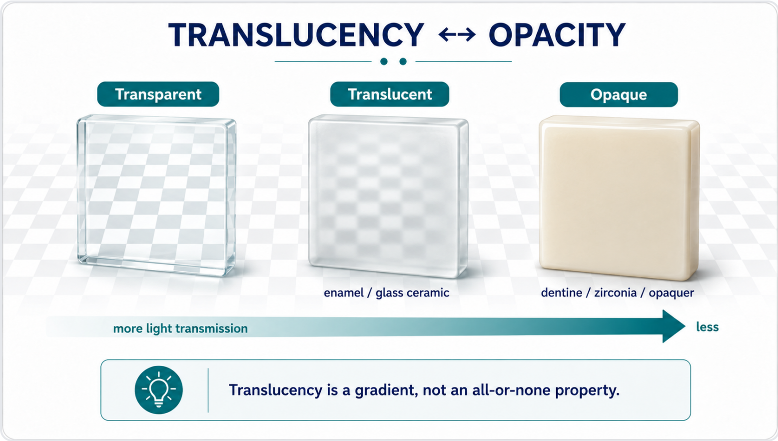
The three attributes of tooth colour.



Value (lightness) is the strongest driver of a match.

Optical properties of teeth & restoratives

- **Translucency / opacity** (translucency parameter, TP): a gradient — enamel more translucent, dentine more opaque.
- **Fluorescence**: teeth absorb UV and re-emit visible light (~440–450 nm); dentine ~3× enamel. A non-fluorescent restoration can look dull/grey under UV (a metameric failure).
- **Opalescence**: enamel scatters short wavelengths → bluish in reflected light, orange in transmitted light (the incisal halo).



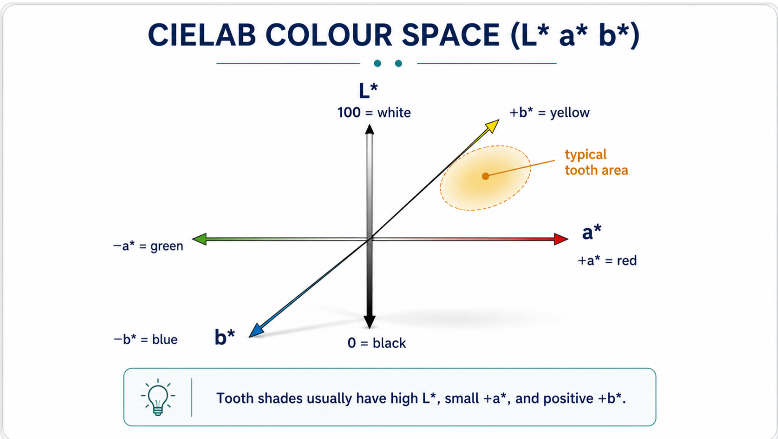
Translucency vs opacity across the tooth.

PERFORMANCE

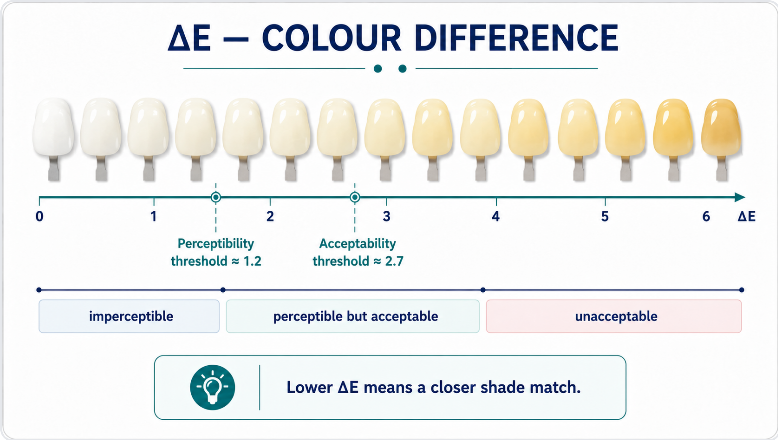
Measuring colour & final appearance

Measuring colour objectively

- **Munsell**: orders colour by Hue Value/Chroma (the basis of Vita-style guides).
- **CIELAB (L* a* b*)**: L* = lightness, a* = green↔red, b* = blue↔yellow — a device-independent 3-D colour space.
- **ΔE** = distance between two colours in that space. Thresholds (Paravina 2015): **perceptibility ΔEab ≈ 1.2**, **acceptability ΔEab ≈ 2.7**.
- **Colorimeter vs spectrophotometer**: the spectrophotometer measures the full reflectance spectrum → more reliable, and helps detect **metamerism** (two colours matching under one light but not another).



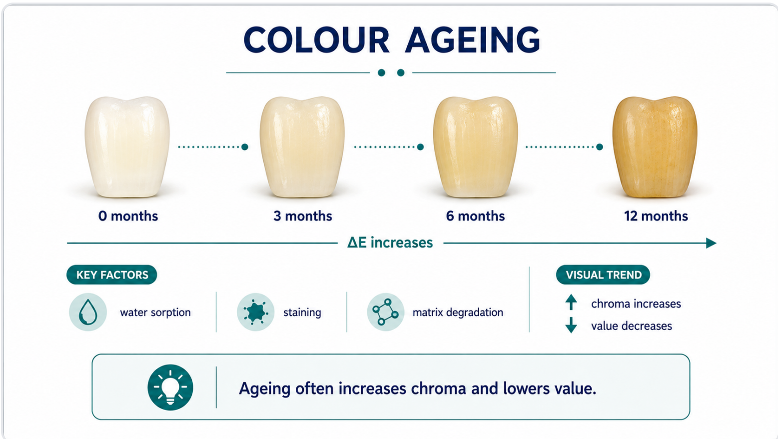
CIELAB: L* (lightness), a* (green–red), b* (blue–yellow).



ΔE thresholds: perceptibility ≈ 1.2, acceptability ≈ 2.7.

What sets the final colour (5 factors) & stability

- **Composition, thickness, background, surface finish, ageing.** Thin/translucent layers let the background show through; polish changes surface reflection.
- **Shade-match conditions matter**: use neutral daylight, match quickly (eye fatigue), remove strong surrounding colours.
- **Ageing**: staining, surface roughening and matrix changes shift colour over time — a stability, not just an initial-match, problem.



Colour is a moving target: ageing and staining shift it over time.

Take-home

- Refractive-index matching (structure) controls translucency; the four fates of light (property) build appearance; value + ΔE + the five factors (performance) decide the clinical match and its stability.

Dental Materials Science · Lecture 04 handout · structure → property → performance.